

Katalog kamera dan handphone dengan fungsi depth sensing

Ringkasan kamera RGB-D dan smartphone yang punya sensor kedalaman, meliputi teknologi stereo, structured light, Time-of-Flight, dan LiDAR.

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13 kamera RGB-D · 11 handphone/tablet

Kamera RGB-D (industri dan robotika)



Microsoft Kinect (v1, for Xbox 360)

Structured light (IR laser projector + monochrome CMOS IR sensor, PrimeSense-based) | Rilis 2010 | \$149.99 at launch (bundled/standalone Xbox 360 peripheral)

Spesifikasi

Depth range ~1.2-3.5 m (near mode ~0.4-3m); depth resolution 640x480 (VGA), 11-bit / 2048 levels; RGB 640x480; FOV 57deg horizontal x 43deg vertical; depth resolution ~0.75mm/px at 50cm, ~3mm/px at 2m

Foto: Wikimedia Commons: <https://commons.wikimedia.org/wiki/File:Xbox-360-Kinect-Standalone.png> (photo by Evan-Amos, 2010) — lisensi: Public domain (released into the public domain by the copyright holder)

Info produk: <https://en.wikipedia.org/wiki/Kinect>



Microsoft Kinect v2 (for Xbox One)

Time-of-Flight (ToF) | Rilis 2013 | ~\$149-199 (standalone 'Kinect for Windows v2' sensor); bundled originally with \$499 Xbox One

Spesifikasi

Depth range ~0.4-4.5 m (mm-level precision); depth resolution 512x424 @ 30fps; RGB 1920x1080; IR stream 512x424; FOV ~70deg x 60deg (also cited as 84.1deg x 53.8deg H/V)

Foto: Wikimedia Commons: <https://commons.wikimedia.org/wiki/File:Xbox-One-Kinect.jpg> (photo by Evan-Amos, 2014) — lisensi: Public domain (released into the public domain by the copyright holder)

Info produk: <https://en.wikipedia.org/wiki/Kinect>

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Orbbec Astra (original series)

Structured light (IR pattern projector + IR camera, PrimeSense-alternative ASIC) | Rilis 2015 | ~\$149-199 at launch (Astra Pro Indiegogo early-bird price was \$99)

Spesifikasi

Depth range ~0.4-8 m depending on variant (Astra Mini S: 0.4-2m; Astra/Astra Pro: 0.6-5m); depth & RGB resolution up to 640x480 @ 30fps (VGA); single USB cable for power+data

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Info produk: <https://www.orbbec.com/products/structured-light-camera/astra-series/>

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Intel RealSense Depth Camera D415

Active/passive stereo IR (IR stereo pair + IR pattern projector, rolling shutter) | Rilis 2018 | ~\$149 at launch (2018); ~\$270 current list price

Spesifikasi

Depth range ~0.5-3 m (ideal); depth resolution up to 1280x720 @ 30fps (up to 1280x720/90fps in some modes); RGB 1920x1080 @ 30fps; depth FOV ~65deg x 40deg (H69.4 x V42.5 RGB FOV); USB 3.1 Gen1 Type-C

Foto: No free/permissive-licensed photo found. Checked Wikimedia Commons (Category: Intel_RealSense has only an L515 lidar photo and a generic RealSense T265 tracking-camera box photo, not D415) and Intel newsroom/press pages (no explicit reusable press-kit image located).

Info produk: <https://www.intelrealsense.com/depth-camera-d415/>

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Intel RealSense Depth Camera D435i

Active stereo IR (global shutter stereo pair + IR projector) plus onboard IMU | Rilis 2019 | ~\$199 at launch (2019); ~\$330 current list price

Spesifikasi

Depth range ~0.3-3 m ideal (0.2-10 m operating); depth resolution up to 1280x720 @ 90fps; RGB 1920x1080 @ 30fps; FOV ~87deg x 58deg; BMI055 6-DoF IMU; USB 3.1 Type-C

Foto: No free/permissive-licensed photo found on Wikimedia Commons or Intel press pages after search.
Info produk: <https://www.intelrealsense.com/depth-camera-d435i/>

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Occipital Structure Sensor (Pro)

Structured light (IR pattern projector), mounts to iPad/iPhone for mobile 3D scanning | Rilis 2019 | \$399-599 depending on bundle

Spesifikasi

Working range ~0.4-3.5 m; accuracy ~+/-0.29% (Pro/healthcare-grade version); mobile/handheld form factor, Lightning/USB-C bracket mount to iOS devices

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Info produk: <https://structure.io/structure-sensor-pro>

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Intel RealSense Depth Camera D455

Active stereo IR (global shutter, wider 95mm baseline) plus IMU | Rilis 2020 | ~\$239 at launch (2020)

Spesifikasi

Depth range ~0.6-6 m; depth resolution up to 1280x720 @ 90fps; RGB 1920x1080 (global shutter); FOV ~86deg x 57deg; depth error <2% at 4m; IMU for motion refinement; USB 3.1

Foto: No free/permissive-licensed photo found on Wikimedia Commons or Intel press pages after search.
Info produk: <https://www.intelrealsense.com/depth-camera-d455/>



Microsoft Azure Kinect DK

Time-of-Flight (ToF) - 1MP AMCW (amplitude-modulated continuous-wave) iToF depth sensor | Rilis 2020 | \$399 at launch

Spesifikasi

Depth range up to 5.46 m (mode-dependent: NFOV ~0.5-3.86m, WFOV ~0.25-2.88m or 2.21-5.46m); depth resolution up to 1024x1024 (NFOV) or 512x512 (WFOV); RGB 12MP; 7-mic circular array; IMU; USB-C. Discontinued Oct 2023, SDK/firmware support through 2027.

Foto: Microsoft News Center (news.microsoft.com) official press photo set for Azure Kinect DK, hosted at msftstories.thesourcemediassets.com. URL verified to resolve (HTTP 200, image/jpeg). — lisensi: Official Microsoft press/media asset intended for editorial/media use. No explicit CC or public-domain license statement was found on the page itself, so treat as a manufacturer press-kit image rather than a confirmed free license; verify Microsoft's press-image terms before redistribution beyond editorial use.
Info produk: <https://azure.microsoft.com/en-us/products/kinect-dk>

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Stereolabs ZED 2i

Passive stereo (dual RGB global-shutter/rolling-shutter pair) with neural-network depth estimation (Neural Depth Engine) | Rilis 2021 | ~\$549-749 depending on lens/IP configuration (exact launch price not publicly confirmed)

Spesifikasi

Depth range up to 20 m (sub-mm accuracy at close range); resolution up to 4416x1242 @ 15fps or 1920x1080 @ 60fps; FOV 110deg (H) x 70deg (V) x 120deg (D); IP66-rated enclosure; 9-DoF IMU; USB 3.0 Type-C

Foto: No free/permissive-licensed photo found on Wikimedia Commons or Stereolabs' press/media pages after search.

Info produk: <https://www.stereolabs.com/store/products/zed-2i>

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Luxonis OAK-D Pro / OAK-D S2

Active stereo IR (stereo pair + IR dot projector + IR illumination LED for low light) with onboard AI (Myriad X VPU) | Rilis 2021 | OAK-D S2 ~\$329; OAK-D Pro ~\$399 (varies by variant: PoE, wide-FOV, etc.)

Spesifikasi

Ideal depth range ~0.7-12 m (Pro); 12MP autofocus/fixed-focus central RGB camera option; 940nm notch-filtered stereo pair; 4 TOPS onboard neural inference; USB3/PoE variants available

Foto: No free/permissive-licensed photo found on Wikimedia Commons or Luxonis press/media pages after search.

Info produk: <https://docs.luxonis.com/hardware/products/OAK-D%20Pro>

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Intel RealSense Depth Camera D457

Active stereo IR plus IMU, industrial GMSL2/FAKRA interface (vs USB on other D400 models) | Rilis 2022 | ~\$520-530

Spesifikasi

Depth range ~0.6-6 m; depth resolution up to 1280x720 @ 90fps; dimensions 124 x 29 x 36 mm; IP65-rated enclosure; GMSL/FAKRA coax connection up to 15 m cable run, supports multi-camera sync over one cable

Foto: No free/permissive-licensed photo found on Wikimedia Commons or Intel press pages after search.

Info produk: <https://www.intelrealsense.com/depth-camera-d457/>

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Orbbec Femto Bolt

Time-of-Flight (ToF) - 1MP iToF sensor, positioned as an Azure Kinect DK replacement (uses Microsoft ToF tech) | Rilis 2023 | Not publicly listed by Orbbec (comparable positioning/tier to the discontinued \$399 Azure Kinect DK)

Spesifikasi

Depth range ~0.25-5.46 m (mode dependent); depth resolution up to 1024x1024 @ 15fps or 640x576 @ 30fps; RGB 4K, 90deg FOV; depth FOV 120deg; std deviation of random error <=17mm; USB-C for power+data

Foto: No free/permissive-licensed photo found on Wikimedia Commons or Orbbec's website/press materials after search.

Info produk: <https://www.orbbec.com/products/tof-camera/femto-bolt/>

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Stereolabs ZED X

Passive stereo (dual 2.3MP global-shutter sensors) with neural depth estimation,
GMSL2 industrial interface | Rilis 2023 | €599 (~\$650)

Spesifikasi

Depth range 0.3-20 m (2.2mm accuracy) or 1-35 m (4mm accuracy); resolution 1200p @ 60fps global shutter; IP67-rated aluminum enclosure; vibration-resistant IMU (16-bit accel+gyro); GMSL2 low-latency interface

Foto: No free/permissive-licensed photo found on Wikimedia Commons or Stereolabs' press/media pages after search.

Info produk: <https://www.stereolabs.com/products/zed-x>

Handphone dan tablet dengan sensor depth

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Oppo R17 Pro / RX17 Pro

Time-of-Flight (ToF) - 3D stereo ToF rear camera | Rilis 2018 | €599 (RX17 Pro, European launch)

Spesifikasi

Triple rear camera: 12MP main (variable aperture) + 20MP secondary + a 3D TOF stereo camera module for AR; among the first Chinese-brand phones with a ToF depth camera (2018)

Foto: No free/permissive-licensed real product photo found on Wikimedia Commons (only a small stylized vector icon render exists, not a real photo) or Oppo's press pages after search.

Info produk: https://www.gsmarena.com/oppo_rx17_pro-9288.php



Samsung Galaxy S10 5G

Time-of-Flight (ToF) - 'DepthVision' 3D depth sensing camera | Rilis 2019 | ~\$1,300 (Verizon exclusive at launch)

Spesifikasi

Quad rear camera including a VGA-class ToF depth sensor for AR mapping and portrait background blur; 6.7in display

Foto: Wikimedia Commons: [https://commons.wikimedia.org/wiki/File:SAMSUNG_Galaxy_S10_5G_\(1\).jpg](https://commons.wikimedia.org/wiki/File:SAMSUNG_Galaxy_S10_5G_(1).jpg) — lisensi: CC BY-SA 4.0

Info produk: https://www.gsmarena.com/samsung_galaxy_s10_5g-9678.php



Samsung Galaxy Note10+

Time-of-Flight (ToF) - 'DepthVision' VGA depth camera | Rilis 2019 | \$1,099.99 at launch

Spesifikasi

Fourth rear camera is a VGA ToF DepthVision sensor enabling 3D scanning/AR measurement and live focus; note the base Note10 (non-Plus) omits the ToF sensor

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Info produk: <https://www.samsung.com/global/galaxy/galaxy-note10/>



Huawei P30 Pro

Time-of-Flight (ToF) rear camera | Rilis 2019 | €999 (128GB) at launch

Spesifikasi

Quad rear camera including a dedicated ToF sensor for AR depth/scale detection and portrait effects; also known for 5x optical / 10x hybrid zoom

Foto: Wikimedia Commons: https://commons.wikimedia.org/wiki/File:Huawei_P30_Pro.jpg — lisensi: CC0 1.0 (public domain dedication)

Info produk: <https://consumer.huawei.com/en/phones/p30-pro/specs/>

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Huawei Mate 30 Pro

Time-of-Flight (ToF) - dual ToF sensors (front and rear) | Rilis 2019 | Not publicly confirmed at global launch (regional pricing only; Huawei's US market exit limited official listings)

Spesifikasi

Quad rear camera with a 3D ToF sensor plus a second 3D ToF sensor on the front camera array, used for depth-based AR, gesture control, and portrait effects

Foto: No free/permissive-licensed photo found on Wikimedia Commons (only Mate 20 Pro / Mate 40 Pro / Mate 50 files exist there) or Huawei's press pages after search.

Info produk: <https://consumer.huawei.com/en/phones/mate30-pro/specs/>

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Honor View20 (V20)

Time-of-Flight (ToF) rear 3D camera | Rilis 2019 | ~\$480 (CNY 3,299, 6+128GB China price) / £499 (UK base model)

Spesifikasi

Dual rear camera: 48MP main (Sony IMX586) plus a 3D ToF sensor used for object/body scanning and AR effects; 6.4in display

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Info produk: https://www.gsmarena.com/honor_view_20-9468.php



LG G8 ThinQ

Time-of-Flight (ToF) - front-facing 'Z Camera' | Rilis 2019 | \$819-900 at launch

Spesifikasi

Front-facing Z Camera ToF sensor used for 'Hand ID' palm-vein authentication and 'Air Motion' touchless gesture control; rear dual camera is conventional (no ToF)

Foto: Wikimedia Commons: [https://commons.wikimedia.org/wiki/File:LG_G8_ThinQ_\(cropped\).png](https://commons.wikimedia.org/wiki/File:LG_G8_ThinQ_(cropped).png) — lisensi: CC BY 2.0

Info produk: https://www.gsmarena.com/lg_g8_thinq-9540.php



Apple iPhone 12 Pro / 13 Pro (and Pro Max variants) - LiDAR

LiDAR (direct Time-of-Flight scanning laser) | Rilis 2020 | \$999 at launch (128GB, iPhone 12 Pro); iPhone 13 Pro also launched at \$999 in 2021

Spesifikasi

LiDAR Scanner range up to ~5 m; used for instant AR plane detection, Night mode portraits, and ARKit depth APIs; iPhone 12 Pro released with 6.1in display, A14 Bionic

Foto: Wikimedia Commons: https://commons.wikimedia.org/wiki/File:IPhone_12_Pro_-_3.jpg — lisensi: CC BY-SA 4.0

Info produk: <https://www.apple.com/newsroom/2020/10/apple-introduces-iphone-12-pro-and-iphone-12-pro-max-with-5g/>



Apple iPad Pro (2020, 4th generation, and later LiDAR models)

LiDAR (direct Time-of-Flight scanning laser) | Rilis 2020 | \$799 (11-inch) / \$999 (12.9-inch) at launch

Spesifikasi

LiDAR Scanner range up to ~5 m; paired with wide + ultra-wide rear cameras; targeted at AR and 3D scanning/measuring apps

Foto: Wikimedia Commons:

https://commons.wikimedia.org/wiki/File:LiDAR_Scanner_and_Back_Camera_of_iPad_Pro_2020_-_3.jpg

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Info produk: <https://www.apple.com/ipad-pro/>



Samsung Galaxy S20+ / S20 Ultra

Time-of-Flight (ToF) - 'DepthVision' 0.3MP depth sensor | Rilis 2020 | \$1,399-\$1,400 at launch (S20 Ultra)

Spesifikasi

Rear quad/penta camera array including a 0.3MP ToF DepthVision sensor for bokeh simulation and AR; note: Samsung dropped the ToF sensor starting with the Galaxy S21 generation, and the S23 Ultra has only a laser-AF rangefinder (not a full ToF depth camera) - verified, so S23 Ultra is excluded from this catalog

Foto: Wikimedia Commons: https://commons.wikimedia.org/wiki/File:Samsung_Galaxy_S20_Ultra.png —

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Info produk: <https://www.samsung.com/global/galaxy/galaxy-s20-ultra/>



Apple iPhone 14 Pro / 15 Pro / 16 Pro (and Pro Max variants) - LiDAR

LiDAR (direct Time-of-Flight scanning laser), same core sensor lineage as iPhone 12/13 Pro with software/ISP refinements | Rilis 2022 | \$999+ at launch across generations (iPhone 14/15/16 Pro)

Spesifikasi

LiDAR Scanner range up to ~5 m; supports ARKit scene depth/mesh APIs, Night mode autofocus, Cinematic mode depth, spatial video (iPhone 15/16 Pro)

Foto: Wikimedia Commons: https://commons.wikimedia.org/wiki/File:IPhone_14_Pro_-_2.jpg — lisensi: CC BY-SA 4.0

Info produk: <https://www.apple.com/iphone-15-pro/>